

P1 习题 6 P25

$$(1+2x-x^2)y'' + (x^2-3)y' + 2(1-x)y = 0$$

$$(1+2x-x^2)e^x + (x^2-3)e^x + 2(1-x)e^x = 0$$

$$(1+2x-x^2+x^2-3+2-2x)e^x = 0$$

$$y_2' = 2ax + b \quad y_2'' = 2a$$

$$(1+2x-x^2)2a + (x^2-3)(2ax+b) + 2(1-x)(ax^2+bx+c) = 0$$

$$\text{展开得} \begin{cases} (b-2b)=0 \\ 4a-ba+2b-2c=0 \\ 2a-3b+2c=0 \end{cases} \rightarrow \text{取} \begin{cases} a=1 \\ b=0 \\ c=-1 \end{cases}$$

$$y_2 = x^2 - 1$$

$$\text{通解: } y = C_1 e^x + C_2 (x^2 - 1)$$

P2. 习题 6 P26 2)

$$x^2 y'' - 2xy' + (x^2 + 2)y = 0 \quad y_1 = x \cos x$$

$$y'' - \frac{2}{x}y' + \frac{x^2+2}{x^2}y = 0$$

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx \\ &= x \cos x \int \frac{1}{x^2 \cos^2 x} e^{\int \frac{2}{x} dx} dx \\ &= x \cos x \int \sec^2 x dx \\ &= x \cos x (\tan x + C) \end{aligned}$$

$$\text{令 } C=0, \quad y_2 = x \sin x$$

$$\text{通解: } y = x(C_1 \cos x + C_2 \sin x)$$

P_{3.4} 习题 6 P₂₇

$$I) x^2(\ln x - 1)y'' - xy' + y = 0$$

$$y_1 = x$$

$$y'' - \frac{1}{x(\ln x - 1)}y' + \frac{1}{x^2(\ln x - 1)}y = 0$$

$$y_2 = x \int \frac{1}{x^2} e^{\int \frac{1}{x(\ln x - 1)} dx} dx$$

$$= x \int \frac{1}{x^2} e^{\ln(\ln(x-1))} dx$$

$$= x \int \frac{\ln(x-1)}{x^2} dx$$

$$= -\ln x + C$$

$$\text{通解: } y = C_1 x + C_2 \ln x$$

$$II). (x-1)y'' - (x+1)y' + 2y = 0$$

$$y_1 = e^x$$

$$y'' - \frac{x+1}{x-1}y' + \frac{2}{x-1}y = 0$$

$$y_2 = e^x \int e^{-2x} e^{\int \frac{x+1}{x-1} dx} dx$$

$$= e^x \int e^{-2x} e^{x+2\ln(x-1)} dx$$

$$= e^x \int (x-1)^2 e^{-x} dx$$

$$= x^2 + 1 + C$$

$$\text{通解: } y = C_1 e^x + C_2 (x^2 + 1)$$

P_{5.6, 7, 8} 习题 6 P₂₇ 1) 4) 5) 6)

$$I) y'' - 3y' + 2y = 0$$

$$\text{特征方程: } r^2 - 3r + 2 = (r-2)(r-1) = 0 \rightarrow r_1 = 1, r_2 = 2$$

$$\text{通解: } y = C_1 e^x + C_2 e^{2x}$$

$$IV) y'' + by' + ay = 0$$

$$\text{特征方程: } r^2 + br + a = (r+3)^2 = 0 \rightarrow r = -3$$

$$\text{通解: } y = (C_1 + C_2 x) e^{-3x}$$

$$V). y'' + by' + 10 = 0$$

$$\text{特征方程: } r^2 + br + 10 = 0 \rightarrow r = \frac{-b \pm 2i}{2} = -3 \pm i$$

$$\text{通解: } y = (C_1 \cos x + C_2 \sin x) e^{-3x}$$

$$VI) y''' - y'' - y' + y = 0$$

$$\text{特征方程 } r^3 - r^2 - r + 1 = (r-1)^2(r+1) \rightarrow r_1 = r_2 = 1, r_3 = -1$$

$$\text{通解: } y = (C_1 + C_2 x) e^x + C_3 e^{-x}$$

P_{9,10} 习题 6 B0 2) 3)

$$II) y'' - 4y' + 3y = 0$$

$$\text{特征方程 } r^2 - 4r + 3 = (r-1)(r-3) = 0 \rightarrow r_1 = 1, r_2 = 3$$

$$\text{通解: } y = C_1 e^x + C_2 e^{3x}$$

$$y' = C_1 e^x + 3C_2 e^{3x}$$

$$\text{代入 } y(0) = 1, y'(0) = 3 \text{ 得 } C_1 = 0, C_2 = 1$$

$$y^* = e^{3x}$$

$$III). 16y'' + 8y' + y = 0$$

$$\text{特征方程: } 16r^2 + 8r + 1 = (4r+1)^2 = 0 \text{ 得 } r = -\frac{1}{4}$$

$$\text{通解: } y = (C_1 + C_2 x) e^{-\frac{1}{4}x}$$

$$y' = C_2 e^{-\frac{1}{4}x} - \frac{1}{4}(C_1 + C_2 x) e^{-\frac{1}{4}x}$$

$$\text{代入 } y(0) = 0, y'(0) = 0.5 \text{ 得 } C_1 = 0, C_2 = \frac{1}{2}$$

$$y^* = \frac{1}{2} x e^{-\frac{1}{2}x}$$

P_{11,12,13} 习题 6 B₁ 2) 3) b)

II) $6y'' - y' - 2y = xe^x$

特征方程: $6r^2 - r - 2 = 0$ $r = \frac{1 \pm 7}{12}$ $\rightarrow$ 基础解系 $e^{\frac{2}{3}x}, e^{-\frac{1}{6}x}$

$6-1-2 \neq 0 \rightarrow 1$ 不是特征根

设 $y^* = (C_1 x + C_2) e^x$

$(y^*)' = (C_1 x + C_1 + C_2) e^x$

$(y^*)'' = (C_1 x + 2C_1 + C_2) e^x$

代回: $6(C_1 x + 2C_1 + C_2) - (C_1 x + C_1 + C_2) - 2(C_1 x + C_2) = x$

$$\begin{cases} 6C_1 - C_1 - 2C_1 = 1 \\ 12C_1 + 6C_2 - C_1 - C_2 - 2C_2 = 0 \end{cases} \rightarrow \begin{cases} C_1 = \frac{1}{3} \\ C_2 = -\frac{11}{9} \end{cases}$$

通解 $y = (\frac{1}{3}x - \frac{11}{9})e^x + C_1 e^{\frac{2}{3}x} + C_2 e^{-\frac{1}{6}x}$

III) $x'' + 4x = t \sin t$

特征方程: $r^2 + 4 = 0 \rightarrow r = \pm 2i \rightarrow$ 基础解系 $\cos 2t, \sin 2t$

$\alpha + i\beta = i$ 不是特征根

设 $x^* = (a_1 t + a_2) \cos t + (b_1 t + b_2) \sin t$

$(x^*)' = a_1 \cos t - (a_1 t + a_2) \sin t + b_1 \sin t + (b_1 t + b_2) \cos t$

$(x^*)'' = -a_1 \sin t - a_1 \sin t - (a_1 t + a_2) \cos t + b_1 \cos t + b_1 \cos t - (b_1 t + b_2) \sin t$

代回: $(-2a_1 + 3b_1 t + 3b_2) \sin t + (3a_1 t + 3a_2 + 2b_1) \cos t = t \sin t$

$$\begin{cases} -2a_1 + 3b_1 t + 3b_2 = t \\ 3a_1 t + 3a_2 + 2b_1 = 0 \end{cases} \rightarrow \begin{cases} a_1 = 0 \\ a_2 = -\frac{2}{9} \end{cases} \begin{cases} b_1 = \frac{1}{3} \\ b_2 = 0 \end{cases}$$

$$\lambda^* = -\frac{2}{9}\cos t + \frac{1}{3}t\sin t$$

通解: $\lambda = -\frac{2}{9}\cos t + \frac{1}{3}t\sin t + C_1\cos 2t + C_2\sin 2t$

VI) $y'' + 2y' + 2y = \lambda + 3 + e^{-\lambda}\sin \lambda$

特征方程 $r^2 + 2r + 2 = 0 \rightarrow r = -1 \pm i \rightarrow$ 基础解系 $e^{-\lambda}\cos \lambda, e^{-\lambda}\sin \lambda$

对于 $y'' + 2y' + 2y = \lambda + 3$

设 $y_1^* = a_1\lambda + a_2$

$(y_1^*)' = a_1 \quad (y_1^*)'' = 0$

代入 $2a_1 + 2a_1\lambda + 2a_2 = \lambda + 3$

$$\begin{cases} 2a_1 = 1 \\ 2a_1 + 2a_2 = 3 \end{cases} \rightarrow \begin{cases} a_1 = \frac{1}{2} \\ a_2 = 1 \end{cases}$$

故 $y_1^* = \frac{1}{2}\lambda + 1$

对于 $y'' + 2y' + 2y = \sin \lambda e^{-\lambda}$

设 $y_2^* = \lambda (A\cos \lambda + B\sin \lambda)e^{-\lambda}$

$(y_2^*)' = (A\cos \lambda + B\sin \lambda)e^{-\lambda} + \lambda(-A\sin \lambda + B\cos \lambda)e^{-\lambda} - \lambda(A\cos \lambda + B\sin \lambda)e^{-\lambda}$
 $= e^{-\lambda} [(A - A\lambda + B\lambda)\cos \lambda + (B - A\lambda + B\lambda)\sin \lambda]$

$(y_2^*)' = -e^{-\lambda} [(A - A\lambda + B\lambda)\cos \lambda + (B - A\lambda + B\lambda)\sin \lambda]$
 $+ e^{-\lambda} [(-A + 2B - A\lambda + B\lambda)\cos \lambda + (B - 2A + A\lambda - B\lambda)\sin \lambda]$
 $= e^{-\lambda} [(-2A + 2B)\cos \lambda + (-2A + 2A\lambda - 2B\lambda)\sin \lambda]$

代入 $\begin{cases} (-2A + 2B + 2A - 2A\lambda + 2B\lambda + 2A\lambda) = 0 \\ -2A + 2A\lambda - 2B\lambda + 2B - 2A\lambda + 2B\lambda + 2B\lambda = 1 \end{cases} \rightarrow \begin{cases} A = -\frac{1}{2} \\ B = 0 \end{cases}$

$y_2^* = -\frac{1}{2}\lambda\cos \lambda e^{-\lambda}$

通解: $y = \frac{1}{2}\lambda - \frac{1}{2}\lambda\cos \lambda e^{-\lambda} + C_1e^{-\lambda}\cos \lambda + C_2e^{-\lambda}\sin \lambda$

P14 习题6 P34

$$f'(x) = e^x + \int_0^x t f(t) dt - \pi \int_0^x f(t) dt$$

$$f'(x) = e^x + \pi f(x) - \int_0^x f(t) dt - \pi f(x) = e^x - \int_0^x f(t) dt$$

$$f'(x) = e^x - f(x)$$

$$f''(x) + f(x) = e^x$$

特征方程 $r^2 + 1 = 0 \rightarrow r = \pm i \rightarrow$ 基础解系 $\cos x, \sin x$

设 $f^*(x) = Ae^x$

$$f'(x)^* = Ae^x \quad f''(x)^* = Ae^x$$

代入 $Ae^x + Ae^x = e^x \rightarrow A = \frac{1}{2}$

$$f^*(x) = \frac{1}{2}e^x$$

通解: $f(x) = \frac{1}{2}e^x + C_1 \cos x + C_2 \sin x$

$$f(0) = 1 \quad f'(0) = 1 \rightarrow \begin{cases} \frac{1}{2} + C_1 = 1 \\ \frac{1}{2} + C_2 = 1 \end{cases} \rightarrow \begin{cases} C_1 = \frac{1}{2} \\ C_2 = \frac{1}{2} \end{cases}$$

$$f(x) = \frac{1}{2}(e^x + \cos x + \sin x)$$

P15 习题6 P35

令 $u = \pi t \rightarrow dt = -\frac{1}{\pi} du$

$$y'(x) = \int_0^x (\pi - u) y(u) du + \sin \pi = \pi \int_0^x y(u) du - \int_0^x u y(u) du + \sin \pi$$

$$y'(x) = \int_0^x y(u) du + \pi y(x) - \pi y(x) + \cos \pi$$

$$y''(x) = y(x) - \sin \pi$$

$$y''(x) - y(x) = -\sin \pi$$

特征方程 $r^2 - 1 = 0 \rightarrow r_1 = \pm 1 \rightarrow$ 基础解系 e^{-x}, e^x

$$\text{设 } y(x)^* = A \sin x + B \cos x$$

$$y'(x)^* = A \cos x - B \sin x \quad y''(x)^* = -A \sin x - B \cos x$$

$$\text{代入 } -A \sin x - B \cos x - A \sin x - B \cos x = -\sin x$$

$$\rightarrow \begin{cases} A = \frac{1}{2} \\ B = 0 \end{cases}$$

$$y(x)^* = \frac{1}{2} \sin x$$

$$\text{通解 } y(x) = \frac{1}{2} \sin x + C_1 e^x + C_2 e^{-x}$$

$$y(0) = 0 \quad y'(0) = 1 \quad \rightarrow \begin{cases} C_1 + C_2 = 0 \\ \frac{1}{2} + C_1 - C_2 = 1 \end{cases} \rightarrow \begin{cases} C_1 = \frac{1}{4} \\ C_2 = -\frac{1}{4} \end{cases}$$

$$y(x) = \frac{1}{2} \sin x + \frac{1}{4}(e^x - e^{-x})$$

P16 习题 6 P36

$$\text{---} \square \rightarrow \infty$$

$$k = \frac{F}{\lambda} = \frac{100}{3} \text{ N/m} \quad \omega = 1.5 \text{ rad/s}$$

$$\omega = \sqrt{\frac{k}{m}} = \frac{10}{3} \text{ rad/s} \quad x = A \cos(\omega t + \varphi)$$

$$v = -A\omega \sin(\omega t + \varphi)$$

$$\text{代入 } x|_{t=0} = 0, v|_{t=0} = 1.5 \text{ 代入}$$

$$\begin{cases} A \cos \varphi = 0 \\ -A\omega \sin \varphi = 1.5 \end{cases} \rightarrow \begin{cases} \varphi = -\frac{\pi}{2} \\ A = 0.45 \text{ m} \end{cases}$$

$$x = 0.45 \cos\left(\frac{10}{3}t - \frac{\pi}{2}\right) = 0.45 \sin\left(\frac{10}{3}t\right) \text{ (m)}$$

P17, 18, 19 习题 6 P37 1) 3) 6)

$$1) x^2 y'' - 3xy' + 3y = 0$$

$$\text{设 } x = e^t \quad y' = \frac{1}{x} \frac{dy}{dt} \quad y'' = \frac{1}{x^2} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right)$$

$$xy' = \frac{dy}{dt}, \quad x^2 y'' = \frac{d^2y}{dt^2} - \frac{dy}{dt}$$

$$[D(D-1) - 3D + 3]y = 0$$

特征方程 $r(r-1) - 3r + 3 = r^2 - 4r + 3 = (r-1)(r-3) = 0 \rightarrow r_1 = 1 \quad r_2 = 3 \rightarrow$ 基本解组 e^t, e^{3t}
 $\downarrow$
 x, x^3

通解: $y = C_1 x + C_2 x^3$

II). $9x^2 y'' + 3xy' + y = 0$

设 $x = e^t$

$$[9D(D-1) + 3D + 1]y = 0$$

特征方程 $9r(r-1) + 3r + 1 = 9r^2 - 6r + 1 = (3r-1)^2 = 0 \rightarrow r = \frac{1}{3} \rightarrow$ 基本解组 $e^{\frac{1}{3}t}, te^{\frac{1}{3}t}$
 $\downarrow$
 $x^{\frac{1}{3}}, x^{\frac{1}{3}} \ln|x|$

通解: $y = x^{\frac{1}{3}} (C_1 + C_2 \ln|x|)$

VI) $(x+1)^2 y'' - 2(x+1)y' + 2y = 0$

令 $x+1 = e^t \quad t = \ln(x+1)$

$$[D(D-1) - 2D + 2]y = 0$$

特征方程 $r(r-1) - 2r + 2 = r^2 - 3r + 2 = (r-2)(r-1) = 0 \rightarrow r_1 = 2 \quad r_2 = 1 \rightarrow$ 基本解组 $e^{2t}, e^t \rightarrow (x+1)^2, x+1$

通解: $y = (x+1) (C_1(x+1) + C_2)$

P₂₀ 补充题 b, P₁₃

I) $f'(x) + f(x) - \frac{1}{x+1} \int_0^x f(t) dt = 0$

$$f'(x) = -f(x) + \frac{1}{x+1} \int_0^x f(t) dt$$

$$f''(x) + f'(x) + \frac{1}{(x+1)^2} \int_0^x f(t) dt - \frac{1}{x+1} f(x) = 0$$

$$f''(x) + f'(x) + \frac{1}{(x+1)} f'(x) = 0$$

$$\frac{df(x)}{dx} = -\frac{x+2}{x+1} f(x)$$

$$\frac{df(x)}{f(x)} = -(1 + \frac{1}{x+1}) dx$$

两边积分 $\ln f(x) = -x - \ln(x+1) + C_1$

$$f(x) = C \frac{e^{-x}}{x+1}$$

代入 $f(0)=1$ $f'(0)+1=0 \rightarrow f'(0)=-1$

$$f'(0) = C = -1$$

$$\text{故 } f(x) = -\frac{e^{-x}}{x+1}$$

II) $x=0$ 时, $f(x)$ 恒成立, 故 $f(x)$ 在 $(0, +\infty)$ 严格递减

$$f(x) \leq 1$$

$$\text{令 } g(x) = f(x) - e^{-x}$$

$$g'(x) = f'(x) + e^{-x} = -\frac{e^{-x}}{x+1} + e^{-x} = e^{-x}(1 - \frac{1}{x+1}) > 0 \text{ 恒成立}$$

故 $g(x)$ 在 $(0, +\infty)$ 上单增.

$$g(x) > g(0) = 1 - 1 = 0 \rightarrow f(x) > e^{-x}$$

综上, $e^{-x} < f(x) \leq 1$ 证.

思考题

P₁

RLC 电路 利用 KVL 有:

$$u(t) = u_R + u_L + u_C$$

$$= R \frac{dq(t)}{dt} + L \frac{d^2q(t)}{dt^2} + \frac{1}{C} q(t)$$

$$\frac{d^2x}{dt^2} + 2\alpha \frac{dx}{dt} + \omega_0^2 x = f(t) \quad (\alpha = \frac{R}{2L}, \omega_0 = \frac{1}{\sqrt{LC}})$$

I) underdamped: $\alpha < \omega_0 \rightarrow$ 振荡 $x(t) = Ae^{-\alpha t} \sin(\omega_d t + \varphi)$

II) overdamped: $\alpha > \omega \rightarrow$ 衰减 $i(t) = A_1 e^{\gamma_1 t} + A_2 e^{\gamma_2 t}$

III). $i(t) = U_m \sin \omega t$, 若 $\omega \approx \omega_0 \rightarrow$ 共振 $I_{max} = \frac{U_m}{\sqrt{(\omega L - \frac{1}{\omega C})^2 + R^2}} = \frac{U_m}{R}$

P2 甲: A_0 B_0 , 增长率: α 移民率: β_1, β_2

$$\frac{dA}{dt} = \alpha A - \beta_1 A + \beta_2 B$$

$$\frac{dB}{dt} = \alpha B - \beta_2 B + \beta_1 A$$

$$\frac{d(A+B)}{dt} = \alpha(A+B)$$

$$\frac{d(A+B)}{(A+B)} = \alpha dt$$

两边积分 $A+B = (A_0+B_0)e^{\alpha t}$

$$\frac{dA}{dt} = \alpha A - \beta_1 A + \beta_2 [(A_0+B_0)e^{\alpha t} - A]$$

$$= A(\alpha - \beta_1 - \beta_2) + \beta_2 (A_0+B_0)e^{\alpha t}$$

$$A' - (\alpha - \beta_1 - \beta_2)A = \beta_2 (A_0+B_0)e^{\alpha t}$$

$$A = e^{\int (\alpha - \beta_1 - \beta_2) dt} \left[\int \beta_2 (A_0+B_0) e^{\alpha t} e^{-\int (\alpha - \beta_1 - \beta_2) dt} dt \right]$$

$$= e^{(\alpha - \beta_1 - \beta_2)t} \left(\int \beta_2 (A_0+B_0) e^{(\beta_1 + \beta_2)t} dt \right)$$

$$= \frac{\beta_2}{\beta_1 + \beta_2} (A_0+B_0) e^{\alpha t} + C e^{(\alpha - \beta_1 - \beta_2)t}$$

代入 $A|_{t=0} = A_0 \rightarrow C = A_0 - \frac{\beta_2}{\beta_1 + \beta_2} (A_0+B_0)$

$$\text{故 } A = \frac{\beta_2}{\beta_1 + \beta_2} (A_0+B_0) e^{\alpha t} + \frac{\beta_1 A_0 - \beta_2 B_0}{\beta_1 + \beta_2} e^{(\alpha - \beta_1 - \beta_2)t}$$

同理 $B = \frac{\beta_1}{\beta_1 + \beta_2} (A_0+B_0) e^{\alpha t} - \frac{\beta_1 A_0 - \beta_2 B_0}{\beta_1 + \beta_2} e^{(\alpha - \beta_1 - \beta_2)t}$